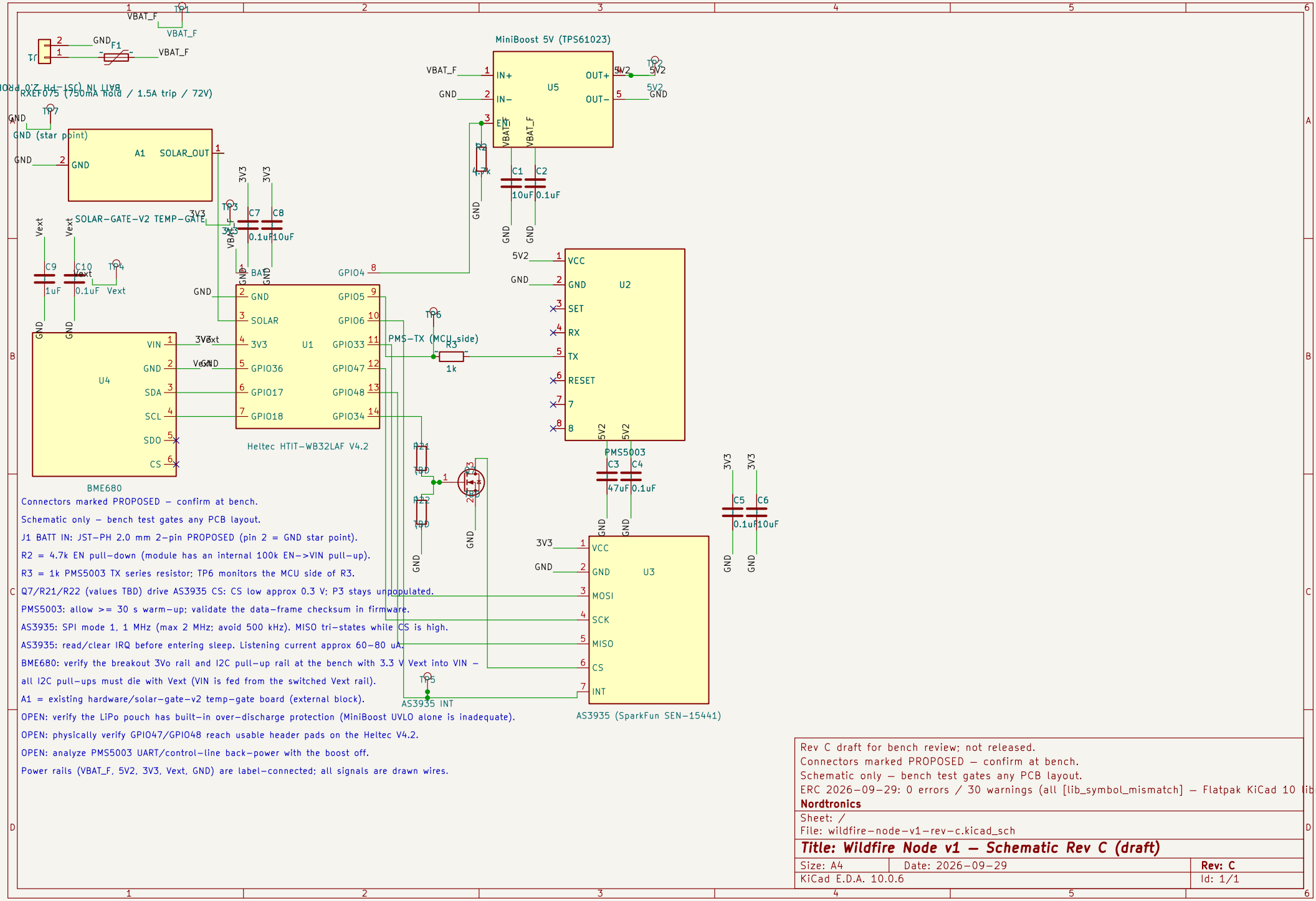


Connectors marked PROPOSED — confirm at bench.

Schematic only — bench test gates any PCB layout.

J1 BATT IN: JST-PH 2.0 mm 2-pin PROPOSED (pin 2 = GND star point).

R2 = 4.7k EN pull-down (module has an internal 100k EN->VIN pull-up).

R3 = 1k PMS5003 TX series resistor; TP6 monitors the MCU side of R3.

Q7/R21/R22 (values TBD) drive AS3935 CS: CS low approx 0.3 V; P3 stays unpopulated.

PMS5003: allow >= 30 s warm-up; validate the data-frame checksum in firmware.

AS3935: SPI mode 1, 1 MHz (max 2 MHz; avoid 500 kHz). MISO tri-states while CS is high.

AS3935: read/clear IRQ before entering sleep. Listening current approx 60-80 uA.

BME680: verify the breakout 3Vo rail and I2C pull-up rail at the bench with 3.3 V Vext into VIN — all I2C pull-ups must die with Vext (VIN is fed from the switched Vext rail).

A1 = existing hardware/solar-gate-v2 temp-gate board (external block).

OPEN: verify the LiPo pouch has built-in over-discharge protection (MiniBoost UVLO alone is inadequate).

OPEN: physically verify GPIO47/GPIO48 reach usable header pads on the Heltec V4.2.

OPEN: analyze PMS5003 UART/control-line back-power with the boost off.

Power rails (VBAT_F, 5V2, 3V3, Vext, GND) are label-connected; all signals are drawn wires.

Rev C draft for bench review; not released.
Connectors marked PROPOSED — confirm at bench.
Schematic only — bench test gates any PCB layout.
ERC 2026-09-29: 0 errors / 30 warnings (all [lib_symbol_mismatch] — Flatpak KiCad 10 libraries vs

Nordtronics

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File: wildfire-node-v1-rev-c.kicad_sch

Title: Wildfire Node v1 — Schematic Rev C (draft)

Size: A4	Date: 2026-09-29	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/1